

Advanced Econometrics for Data Science – Lecture 07

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Agenda

- ① Quick tour de tobit
- ② Tobit
 - a) *corner solutions*
 - b) *censoring data*
- ③ Truncated regression

Quick tour de tobit

Tobin's model - Tobit

Tobit model is used if the dependent variable is limited to a certain interval. What is more, we observe many observations for which the dependent variable takes value equal to one of the ends of the interval.

We can distinguish two mechanisms of generating such variables:

- censoring
- corner solutions

Estimation is performed the same way, but different is the way of parameters' interpretation.

Censored variable

A variable is censored if we observe its values if they are in some interval.

Let's say that the interval is $[a, b]$.

- If the variable is lower than a , then we observe a as a value.
- Similarly, if the variable is greater than b , then we observe b .

Example 14.2 [Mycielski] - respondent's income question

Corner solution

For many samples we observe positive values of the dependent variable and a significant amount of zeros.

Example: Insurance expenditures

Zeros are due to rational decisions of economic entities.

If a variable may takes non-negative values, then it may happen that optimisation process can lead to the verge of the interval of possible solutions. Such solution is called a corner solution.

Tobit in R

- Fair, R.C. (1978). A Theory of Extramarital Affairs. *Journal of Political Economy*, 86, 45–61
- A data frame containing 601 observations on 9 variables.
- **affairs** numeric. How often engaged in extramarital sexual intercourse during the past year? 0 = none, 1 = once, 2 = twice, 3 = 3 times, 7 = 4–10 times, 12 = monthly, 12 = weekly, 12 = daily.
- **gender** factor indicating gender.
- **age** numeric variable coding age in years: 17.5 = under 20, 22 = 20–24, 27 = 25–29, 32 = 30–34, 37 = 35–39, 42 = 40–44, 47 = 45–49, 52 = 50–54, 57 = 55 or over.
- **yearsmarried** numeric variable coding number of years married: 0.125 = 3 months or less, 0.417 = 4–6 months, 0.75 = 6 months–1 year, 1.5 = 1–2 years, 4 = 3–5 years, 7 = 6–8 years, 10 = 9–11 years, 15 = 12 or more years.
- **children** factor. Are there children in the marriage?

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- **religiousness** numeric variable coding religiousness: 1 = anti, 2 = not at all, 3 = slightly, 4 = somewhat, 5 = very.
- **education** numeric variable coding level of education: 9 = grade school, 12 = high school graduate, 14 = some college, 16 = college graduate, 17 = some graduate work, 18 = master's degree, 20 = Ph.D., M.D., or other advanced degree.
- **occupation** numeric variable coding occupation according to Hollingshead classification (reverse numbering).
- **rating** numeric variable coding self rating of marriage: 1 = very unhappy, 2 = somewhat unhappy, 3 = average, 4 = happier than average, 5 = very happy.

Tobit in R

Call:

```
censReg(formula = affairs ~ age + yearsmarried + religiousness +  
         occupation + rating, left = 0, right = Inf, data = Affairs)
```

Observations:

Total	Left-censored	Uncensored	Right-censored
601	451	150	0

Coefficients:

	Estimate	Std. error	t value	Pr(> t)	
(Intercept)	8.17420	2.74145	2.982	0.00287	**
age	-0.17933	0.07909	-2.267	0.02337	*
yearsmarried	0.55414	0.13452	4.119	0.0000379755	***
religiousness	-1.68622	0.40375	-4.176	0.0000296183	***
occupation	0.32605	0.25442	1.282	0.20001	
rating	-2.28497	0.40783	-5.603	0.0000000211	***
logSigma	2.10986	0.06710	31.444	< 0.00000000000000002	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

tobit_marginal_effects.R

```
> tobit_marginal_effects(tobit1, x, dummies_indices=c())
```

Marginal effects of the tobit model

```
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```

	y*	E(y x)	E(y x,y>0)	Pr(y>0 x)	at X=
age	-0.1945103	-0.04938095	-0.04734533	-0.007553498	32.487521
yearsmarried	0.5764621	0.14634830	0.14031540	0.022385991	8.177696
religiousness	-1.6953493	-0.43040380	-0.41266132	-0.065836196	3.116473
female!	-1.0492278	-0.26637086	-0.25539028	-0.040745096	0.000000
occupation	0.1971279	0.05004549	0.04798247	0.007655148	4.194676
rating	-2.2874235	-0.58071560	-0.55677684	-0.088828459	3.931780

(!) indicates marginal effect was calculated for discrete change of dummy variable from 0 to 1

The end of
Quick tour de tobit

Example 14.1 Censoring - stock market example

There is a limitation of a daily price change $\pm 10\%$ (Warsaw stock exchange). If equilibrium is set with greater price change, then quotations are suspended and the price change is $\pm 10\%$. If we observe price change equal to $+10\%$, then there was a change greater than $+10\%$.

Example 14.2 Censoring – question on respondent's income

In our survey we would like to ask "How much do you earn?". We expect that those of high income are likely to omit the question. Therefore, our survey contains:

- 1 Do you earn more than 10K a month? (if so, skip question 2)?
- 2 How much do you earn?
- 3 ...

Football (soccer) matches ticket demand on the basis of the stadium with highest number of spectators in Europe. Which one?

Corner solution or censored case?

- ① Alcohol expenditures
- ② Respondents' income question
- ③ Demand for concert ticket
- ④ In September of 1979, the National Highway Traffic Safety Administration (NHTSA) passed a bill which stated that all car, pickup truck and motorcycle speedometers were to display a maximum speed of 85 miles-per-hour. Thus, a car moving faster than 85 miles-per-hour would have had 85 at its speedometer.
- ⑤ A driver of a car which speed was limited to 110 kilometers-per-hour. We are interested in factors that affects speed.
- ⑥ Variable: fixed assets expenditures divided by total income
- ⑦ Number of hours in an additional workplace of male head of families

Marginal effects

The effect of an additional unit of x variable on probability of a non-zero value of the dependent variable y

$$\frac{\partial \Pr(y > 0|x)}{\partial x} \quad (1)$$

The effect of an additional unit of x variable on the expected value of y for observations with non-zero values of y

$$\frac{\partial \mathbb{E}(y|x, y > 0)}{\partial x} \quad (2)$$

The effect of an additional unit of x variable on the expected value of y

$$\frac{\partial \mathbb{E}(y|x)}{\partial x} \quad (3)$$

tobit_marginal_effects.R

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> tobit_marginal_effects(tobit1, x, dummies_indices=c())
```

Marginal effects of the tobit model

```
-----
```

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age	-0.1945103	-0.04938095	-0.04734533	-0.007553498	32.487521
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(!) indicates marginal effect was calculated for discrete change of dummy variable from 0 to 1

Tobit model assumptions

$$\begin{aligned}y_i^* &= \mathbf{x}\boldsymbol{\beta} + \boldsymbol{\varepsilon}_i \\ \boldsymbol{\varepsilon}_i &\sim N(0, \sigma^2) \\ y_i &= \begin{cases} y_i = y_i^* & \text{for } y_i^* \geq 0 \\ y_i = 0 & \text{for } y_i^* < 0 \end{cases} \end{aligned} \tag{4}$$

and all observations are independent of each other.

Censoring data example – Signal Iduna Park Attendance

Tobit estimates

Log likelihood = -61.56519

Number of obs	=	49
LR chi2(9)	=	21.38
Prob > chi2	=	0.0111
Pseudo R2	=	0.1479

Attendance	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
bvbpos	.0149135	.9260445	0.02	0.987	-1.856692	1.886519
guestspos	.0591823	.27508	0.22	0.831	-.496775	.6151397
monday	-25.63684	8.911355	-2.88	0.006	-43.64736	-7.626325
saturday	2.70568	5.120852	0.53	0.600	-7.643948	13.05531
sunday	4.385378	5.472862	0.80	0.428	-6.675689	15.44644
tuesday	42.21546	.	.	.	.	.
spring	9.062023	4.618412	1.96	0.057	-.2721357	18.39618
summer	1.2285	5.248127	0.23	0.816	-9.37836	11.83536
autumn	1.732069	3.56175	0.49	0.629	-5.466497	8.930635
_cons	79.43356	7.109038	11.17	0.000	65.06566	93.80146
_se	6.779345	1.326254	(Ancillary parameter)			

Obs. summary: 15 uncensored observations
 34 right-censored observations at Attend~e>=81.36

Theorem 19.1 Density of a Truncated Random Variable

If a continuous random variable x has a pdf $f(x)$ and a is a constant, then

$$f(x|x > a) = \frac{f(x)}{1 - \mathbb{P}(X > a)}. \quad (5)$$

Censored Normal Distribution

$$y_i = \begin{cases} y_i = y_i^* & \text{dla } y_i^* \geq 0 \\ y_i = 0 & \text{dla } y_i^* < 0 \end{cases} \quad (6)$$

The distribution that applies if $y^* \sim N[0, \sigma^2]$ is

$$\mathbb{P}(y_i^* < 0) = F(0) = \Phi(0) \quad (7)$$

$$\mathbb{P}(y_i^* = z | y_i^* > 0) = \phi\left(\frac{z}{\sigma}\right) \quad (8)$$

Truncated regression

- Truncation and censoring are two distinct phenomena that cause our samples to be incomplete.
- If we ignore truncation or censoring when analysing our data, our estimates of population parameters will be inconsistent.
- Truncation or censoring happens during the sampling process. Let's begin by defining left-truncation and left-censoring:
- Our data are left-truncated when individuals below a threshold are not present in the sample. For example, if we want to study the size of certain fish based on the specimens captured with a net, fish smaller than the net grid won't be present in our sample.

Source: <https://blog.stata.com/> (link)

Truncated regression

- <https://stats.idre.ucla.edu/stata/output/truncated-regression/>
- Examples of truncated regression
- Example 1.
- A study of students in a special GATE (gifted and talented education) program wishes to model achievement as a function of language skills and the type of program in which the student is currently enrolled. A major concern is that students are required to have a minimum achievement score of 40 to enter the special program. Thus, the sample is truncated at an achievement score of 40.
- Let's estimate this model in R

Truncated regression in R

Call:

```
truncreg::truncreg(formula = achievement ~ lang.score + program2 +  
  program3, data = truncdata, point = 40, direction = "left")
```

BFGS maximization method

35 iterations, 0h:0m:0s

$g'(-H)^{-1}g = 3.97E-10$

Coefficients :

	Estimate	Std. Error	t-value	Pr(> t)	
(Intercept)	9.889564	0.683552	14.4679	< 0.000000000000000022	***
lang.score	0.701698	0.012003	58.4615	< 0.000000000000000022	***
program2TRUE	3.780889	0.221834	17.0438	< 0.000000000000000022	***
program3TRUE	-1.224689	0.219914	-5.5689	0.00000002563	***
sigma	1.196063	0.064649	18.5009	< 0.000000000000000022	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log-Likelihood: -280.54 on 5 Df

Truncated or censored?

- A different problem
- Let's assume that we would like to estimate a model for the results of the final high-school exam, i.e. A-level exam or SAT (Scholastic Aptitude Test) results
- Let's assume that the results are 0-100
- Is this a case of censoring or truncation?

Censored and truncated data

